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CNReader: a reading practice tool designed for Chinese children with developmental dyslexia

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Developmental dyslexia among Chinese-speaking children presents distinct linguistic, cognitive, and information-processing deficiencies, negatively affecting academic performance and cognitive development. Existing dyslexia interventions largely rely on manual guidance, and due to limited resources and a narrow practice approach, results often fall short. This paper presents an innovative tool, CNReader, designed to support reading training for young Chinese children with dyslexia. Incorporating user-friendly visual text presentation, personalized reading guidance, and artificial intelligence (AI) reading support, CNReader aims to stimulate and enhance children's reading interest and ability. Our study engaged children with mild dyslexia across three experimental phases. Results demonstrated that CNReader significantly reduced reading error rates, enhanced reading fluency, and improved comprehension. Notably, AI-paired reading emerged as both more appealing to children and more effective than conventional human-assisted approaches. Furthermore, following a 1-month training period, participants exhibited marked improvements in reading speed, accuracy, and engagement. This research validates CNReader's efficacy as an intervention tool for Chinese dyslexia and underscores the potential of AI integration in reading assistance technologies designed for children with reading disabilities.

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Introduction

Despite having normal intelligence, motivation, and basic education, children with dyslexia underperform in reading fluency and accuracy relative to their peers (Chen et al., 2018). About 8.0% of children in China reportedly struggle with reading and writing Chinese characters and words (Liu et al., 2013; Zhang et al., 2023). While dyslexia can be alleviated with suitable interventions (Franceschini et al., 2013; Helenius et al., 1999; Lin et al., 2013), training and intervention for children with dyslexia in Chinese face multiple challenges. These include insufficient and uneven distribution of professional training, a limited number of teachers and mental health professionals, and difficulties in providing targeted support (Ye, 2023). Moreover, educational resources are scarce, with materials and training tools not designed to meet specific needs, making it challenging to create environments conducive to children with dyslexia (Ye, 2023). The cost of personalized treatment is high, imposing financial burdens on families, and the expense of teacher training further exacerbates the issue. These problems hinder the effective intervention and long-term development of children with dyslexia.

Unlike the phonological processing and decoding difficulties commonly observed in alphabetic languages, the logographic nature of Chinese characters presents these children with unique challenges. Research indicates that children with Chinese dyslexia typically exhibit deficits in phonological awareness, morphological awareness, orthographic awareness, and rapid automatized naming (RAN) (Lin, 2021; Huang et al., 2023). Phonological awareness is crucial for decoding and understanding the sound structure of language, an area in which these children often struggle. Morphological awareness affects their ability to discern and use the smallest units of meaning in the language, whereas orthographic awareness involves recognizing visual patterns and the rules within the writing system. RAN deficits impede the swift retrieval of linguistic information, further hindering reading fluency. Mastery of prosody and rhythm significantly enhances children's reading comprehension and memory in Chinese-reading training (Li et al., 2023; Chen and Wang, 2019). In addition, spatial attention and sustained attention are key components of the reading process, which are often impaired in children with reading difficulties. Studies have shown that spatial attention deficits in children with dyslexia may be associated with comorbid conditions such as attention deficit hyperactivity disorder, further complicating the reading process (Zhang, 2021; Deng, 2023).

Formative research conducted with 13 dyslexic children and their parents revealed the main difficulties faced by these children when learning Chinese characters in first and second grades, which include character omission during reading, poor reading rhythm and prosody, reading errors, and limited sentence and passage comprehension. Building upon previous theoretical research, we introduce CNReader, an intelligent reading application. CNReader offers a user interface designed to be friendly for individuals with reading difficulties, real-time visual prompts, sentence segmentation assistance, an AI companion for reading, and the generation of images related to the reading text to aid comprehension. By integrating enhanced speech processing, morphological understanding, and orthographic recognition capabilities, CNReader aims to alleviate the reading challenges faced by children with Chinese-reading difficulties. CNReader employs visually presented text with color-coded cues and structured layouts to assist children in more effectively decoding and recognizing characters, thereby enhancing orthographic awareness. The auxiliary sentence segmentation feature provides guidance in reading rhythm and prosody, aiding in the

improvement of phonological and morphological awareness. Additionally, visual cues and interactive speech-assisted elements help direct the child's attention through the text, thereby reducing spatial neglect. The AI-paired reading mode offers personalized practice with varying reading complexities and text lengths, training sustained attention, and gradually enhancing the child's ability to maintain focus over extended periods.

We performed three experiments to assess CNReader's effectiveness and user experience. In Experiment 1, 75 dyslexic children were involved to test CNReader's single reading mode while comparing reading accuracy, rhythm, and semantic comprehension between the experimental and control groups. Experiment 2 evaluated CNReader's AI-paired reading feature involving 44 dyslexic children, comparing it with human-paired reading. Experiment 3 investigated CNReader's medium and long-term effects, involving 15 dyslexic children and their parents in a month-long endeavor to gauge the children's reading levels and experiences post a month of CNReader training and their parents' feedback. The main contributions of this paper include:

Put forward the design implications of Chinese dyslexia practice tools through formative research.

The design of an accessible reading practice tool, CNReader, for children with mild Chinese dyslexia.

Experimental evaluation of CNReader's effectiveness and user experience.

Related work

Chinese dyslexia. There is a general consensus among researchers that dyslexia in Chinese reading is associated with deficiencies in four cognitive language skills, despite divergent views on its causes. These are poor phonological awareness, morphological awareness, poor understanding of the visual orthography of Chinese characters, and poor performance in RAN tasks (McBride et al., 2018; Huang, 2022; Vellutino et al., 2004). As per the 2022 version of "Compulsory Education: Chinese Curriculum Standards" in China, students in the first and second grades are required to recognize 1600 Chinese characters, and of these, 800 should be writable (Ministry of Education of the People's Republic of China, 2022). However, these students' Chinese character recognition ability falls significantly short of this educational benchmark. Experimental data reveal that dyslexic students read a single Chinese character an average of 94 ms slower than their age-matched peers. This delay averages 48 ms when reading a single number (Meng, 2018).

Numerous research studies and various training methods have been explored to improve dyslexia, focusing on four basic principles: providing successful experiences, feedback, ample training, and individualized learning steps (Meng, 2018). Mainstream intervention strategies primarily target physiological and cognitive aspects, with physiological interventions often utilizing clinical strategies such as auditory correction and speech teaching (Liu et al., 2013). Comprehensive intervention programs and training methods encompassing oral language, visual orthography knowledge, morphological awareness, and character recognition are widely applied to address cognitive issues (Fung et al., 2022; Chung and Ho, 2010; Signor et al., 2020; Zhang and Sun, 2016). In essence, Chinese dyslexic children need to concentrate on enhancing and training areas like syllable and phoneme recognition, vocabulary comprehension, reading fluency, reading strategies, and fostering reading interest. Nevertheless, there remains a scarcity of universally accepted and effective reading training aids for children with reading disorders in simplified Chinese.

Design for children with dyslexia. Dyslexic children commonly experience difficulties with word recognition, spelling, and decoding (International Dyslexia Association, 2021; Saunier et al., 2022; Yong and Asmuri, 2021). The advancements in technology, especially in the field of human-computer interaction design, have proven effective in assisting these children. Several interactive tools like reading aids, training games, and user interfaces have been developed to strengthen children's reading skills (Gupta et al., 2022; Wu et al., 2019; Wessel et al., 2021). These tools employ auditory, visual, and tactile stimuli to aid in knowledge comprehension and retention. For example, Cosmic Sounds (Brennan et al., 2022) improves phonological awareness through word games. PhonoBlocks (Fan et al., 2016) provides a multisensory learning environment using physical objects. Moreover, games like Character Alive (Fan et al., 2019) use color cues, radical cards, and animations to nurture Chinese literacy skills. Tools created by Osman and Uda (2014) assist in understanding word meanings through innovative visual cues like noun-specific images and verb-specific GIFs. Apps like Augmentally (Gupta et al., 2021) augment reading experiences by scanning textbooks and translating the content into a customizable interface.

Despite advancements, current research exhibits notable limitations. First, existing intervention designs lack sophisticated personalization mechanisms (Fan et al., 2019, 2024). Children with reading disabilities present heterogeneous cognitive profiles, yet prevailing methodologies typically employ standardized approaches that cannot dynamically adapt to individual disability characteristics. Second, current dyslexia intervention tools predominantly need continuous professional human supervision. Although AI has demonstrated considerable efficacy as educational support in various contexts (Denny et al., 2024; Liu et al., 2024), children with dyslexia have not fully benefited from these technological affordances—highlighting a disparity between technological potential and practical implementation for this vulnerable population. Addressing these challenges, this research aims to develop a comprehensive, AI-driven solution specifically designed for Chinese children with developmental dyslexia, emphasizing personalized reading interventions and intelligent reading companion systems.

Assistive technology for dyslexia. Computer-assisted learning can provide a multisensory experience tailored to a variety of learning styles. For dyslexic individuals, technology can offer auditory input via text-to-speech functionality, and visual enhancement through graphics and animations to supplement written content (Sidhu and Manzura, 2010; Powell et al., 2004). Assistive technologies (ATs) for dyslexia provide reading assistance through personalized parameters, including text customization, simplification, text-to-speech synthesis (Haug and Klein, 2018), combined with scanning, Optical Character Recognition technology, character recognition technology, natural language processing technology, and visual recognition technology (Rauschenberger et al., 2019; de Avelar et al., 2015; Fourney et al., 2018; Mohana et al., 2023).

ATs offer reading support and learning opportunities for children with dyslexia (Rauschenberger et al., 2019; Huss et al., 2011; Madeira et al., 2015); however, the logographic nature and polysemic characteristics of Chinese characters present unique challenges that conventional approaches inadequately address. AI technology, with its advanced pattern recognition and semantic processing capabilities, proves particularly suitable for managing the morphological complexities and semantic relationships inherent in Chinese characters. Deep learning algorithms effectively discriminate between visually similar characters (e.g.,

“己”/“已,” “土”/“士”) and resolve ambiguities through contextual analysis. Furthermore, AI systems can monitor reading behaviors in real-time, identify specific difficulty patterns, and deliver targeted interventions (Goodman et al., 2022) while adaptively modifying content difficulty based on performance metrics to create individualized learning trajectories. This research leverages these AI advantages by integrating scanning recognition, natural language processing, speech synthesis, and intelligent interaction into the CNReader system, establishing a comprehensive framework designed to provide precise, effective support for children with Chinese dyslexia.

Formative research

Methodology. In order to gain a deeper understanding of the specific difficulties faced by children with mild Chinese dyslexia during reading, and to determine the design objectives, we conducted research including reading tests, observations, and parental interviews. The study was conducted with 13 first graders with mild dyslexia, all of whom participated in the experiment with their parents' consent. The test materials were selected from a text in the first-grade elementary school curriculum. During the observation process, we noted issues such as omissions, additions, misreading, sentence fragmentation, and difficulty in semantic comprehension. Meanwhile, parental interviews revealed their helplessness towards their children's reading difficulties due to the lack of professional assistance, suitable materials for dyslexic children, and their inability to provide adequate tutoring due to work commitments. Some parents also expressed that their children's academic performance was affected by dyslexia, but the children were unable to receive adequate tutoring from their teachers.

Key findings and design implications. Based on the above research findings, we set the following design goals for reading training tools for children with Chinese dyslexia:

1. User-friendly reading interface: Design the textual format based on the reading preferences of children with dyslexia and provide them with high-quality reading training materials.
2. Real-time reading feedback: Provide real-time feedback on children's reading, marking already read content, and reading errors.
3. Personalized reading guidance: Set the speed of visual guidance for reading based on individual needs, and provide sentence-breaking hints for long and difficult sentences to optimize reading rhythm.
4. Diversified reading training methods: In addition to conventional training methods, provide innovative reading training modes such as human-intelligence interaction, reading challenges, and automatic generation of images from text content, making reading training more attractive.

Design of CNReader

Based on the design objectives outlined above, we carried out the system design for CNReader. This section primarily introduces the interface design and the reading practice mode design of CNReader.

Dyslexia-friendly interface. Drawing on the research of scholars at home and abroad on dyslexia-friendly text formats, we designed the system interface of CNReader (Rello et al., 2012; Rello and Yates, 2014; Yang, 2019a, 2019b). As shown in Fig. 1a, we adopted visually friendly black (000000) typeface on a cream-colored (FAFAC8) background (Madeira, 2015), and set the font

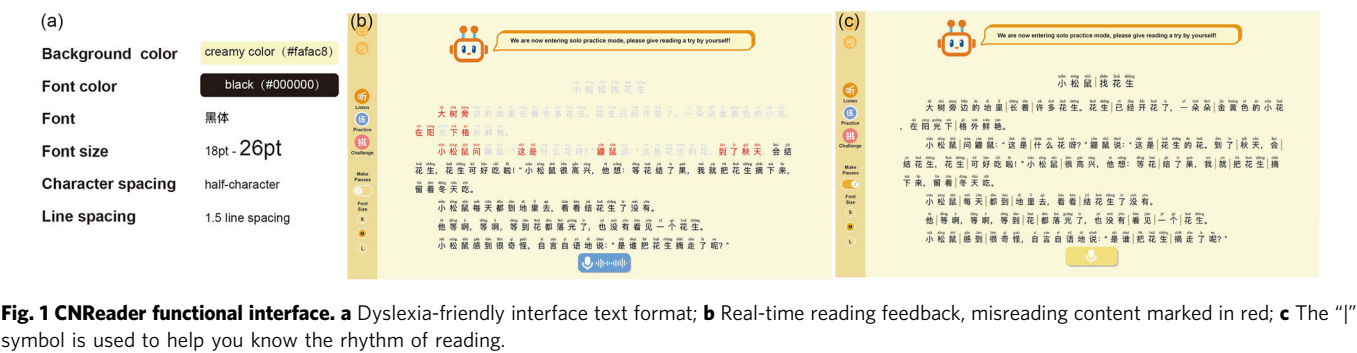


Fig. 2 AI-paired reading mode. **a** Interface of AI-paired reading; **b** The AI voice assistant stops reading at random, and children need to relay the reading.

size between 18 and 26 for faster reading speed (Rello et al., 2013). We used a Heiti font, which is friendlier to dyslexics (Yang, 2019a, 2019b). Studies have found that the narrower the spacing between characters and lines, the longer it takes to read an article. Combining this with user preferences when using the prototype on mobile devices, CNReader chose a half-character spacing and 1.5 line spacing (Yang, 2019a, 2019b). We also set up buttons to allow for personalized adjustment of font size and spacing, which children can adjust based on their personal preferences. Furthermore, CNReader has a text format conversion function, which recognizes regular text and converts it into a dyslexia-friendly mode.

Single-player reading practice mode. The CNReader’s single-player practice mode facilitates repeated practice for children. Children can press the button to listen to the system reading, with the font color changing in real-time to denote the content that has been read, allowing them to keep pace. They can also choose to read independently, with the system providing real-time evaluation of their reading content, as shown in Fig. 1b, with incorrect content highlighted in red. Sentence segmentation, a fundamental knowledge and skill in primary school Chinese classes, is learned by the children in grades 1–2. To cater to actual Chinese-reading needs, the CNReader offers a sentence segmentation aid, as illustrated in Fig. 1c. The symbol “|” is employed to segment long sentences, enabling children to analyze sentence structures clearly, comprehend the meaning accurately, and enhance their reading skills. Children can choose to enable or disable this feature based on their individual needs during practice.

AI-paired reading mode. Paired reading is a tutoring strategy specially designed for beginning readers and children with reading difficulties. Research has found that children participating in a paired reading program for 10–15 min a day, 4–5 times a

week, for a total of 7 weeks (which can be extended) significantly improve their vocabulary recognition, reading fluency, and comprehension skills (Meng, 2018). In response to this, we designed an AI voice assistant for CNReader, whose primary function is to read with the children as a partner.

The current AI-paired reading mode is relay reading, where the voice assistant starts reading and randomly stops, and then the child is required to continue reading the assigned part (Fig. 2a, b). CNReader implements adaptive learning algorithms capable of capturing children’s pronunciation and fluency, including identifying subtle errors and providing specific feedback to correct them. Additionally, it can adjust the AI reading difficulty and speed based on a child’s progress and individual needs, dynamically altering the reading speed and the difficulty of the reading material.

Additionally, the voice assistant aims to increase the children’s engagement. The voice agent’s response comprises two parts: Positive utterances and context-based utterances. Positive utterances are brief verbal responses acknowledging what the children have said, such as “good,” “keep it up,” and “excellent.” Context-based utterances provide feedback on the children’s reading results, like “You made two mistakes, try again!” or “It would be great if your reading was smoother!” CNReader also integrates affective computing technology to assess and respond to the emotional states of children. For instance, if a child exhibits signs of frustration or boredom during reading, the voice assistant can introduce motivational statements or suggest brief interactive breaks.

Reading challenge mode. To enrich the methods of reading training and provide an engaging reading experience, CNReader offers three different levels of reading challenge modes. As shown in Fig. 3a, in Challenge Level 1, sentences in the reading material are not segmented. Challenge success is confirmed when all words



Fig. 3 Reading challenge mode. **a** Reading difficulty setting; **b** After the challenge is completed, the system can generate pictures according to the content of the article as a reward, and the child can download and save.

are read correctly; in Challenge Level 2, without sentence segmentation, the system provides a visual guide at a normal reading speed, requiring children to keep up with the pace, and to read the words correctly; in Challenge Level 3, children are required to select the image related to the reading content from several generated images. Upon successful completion of all three levels, a reward in the form of an image related to the content of the article is unlocked (Fig. 3b). This serves to aid in the comprehension of the article’s content.

Studies

To test the effectiveness and user experience of CNReader, we conducted three studies. Study 1 tested the effectiveness of CNReader’s single-player reading practice mode. Study 2 tested the functionality of CNReader’s AI partner reading mode, and Study 3 tested the medium-long-term practice effect and user experience of CNReader. We focus on intervention for Chinese-reading children with reading disabilities in lower grades, as this period represents a critical window for Chinese character recognition and reading development. Early intervention can effectively correct inappropriate reading patterns and establish suitable strategies, thereby mitigating the negative impacts of reading difficulties on children’s academic achievement and psychological development (Wang, 2024).

The children participating in the studies all have mild reading disorders and have undergone Chinese character recognition tests to assess their language skills. Specifically, the assessments included tests on Chinese character recognition accuracy and speed, as well as evaluations of elementary school students’ reading speed and comprehension level (Meng, 2000, 2018). After testing, the total scores of each assessment indicator were converted into composite scores with a mean of 100 and a standard deviation of 15, resulting in the Reading Quotient (RQ) (Meng, 2000, 2018). An RQ between 90 and 110 indicates typical developmental levels, 110 to 120 is above average, and above 120 is considered excellent. An RQ below 90 suggests the need for improvement and appropriate intervention (Meng, 2000, 2018). In addition, the children also undertook Raven’s Standard Progressive Matrices to assess their non-verbal intelligence. Test results between 90 and 110 indicate normal intelligence.

Study 1

Participants. This study recruited a total of 75 first-grade children with mild dyslexia from elementary schools in Jiujiang, China. The recruitment of participants was facilitated by Chinese

Table 1 Background information about the participants of study 1.				
Group	N	Age	RQ	SPM
Baseline	25 (F = 13, M = 12)	M = 6.60, SD = 0.49	84.04	97.40
CNReader	25 (F = 11, M = 14)	M = 6.88 SD = 0.71	83.08	96.60
X app	25 (F = 11, M = 14)	M = 6.64 SD = 0.56	84.36	99.08

language teachers and class supervisors from the respective schools. Prior to the commencement of the study, researchers thoroughly explained the purpose, procedures, potential risks, and benefits of the study to the children and their parents. It was explicitly communicated that participation was entirely voluntary, and individuals could withdraw at any time without any repercussions. All participants and their parents signed informed consent forms. To ensure the privacy and confidentiality of participant data, all personal information was anonymized and securely stored. Additionally, multiple measures were taken to safeguard the well-being of the children during the research process. This included providing support from school counselors and psychologists to promptly address any signs of discomfort or distress. Through these measures, we aim to uphold high standards of research ethics and ensure a safe and respectful research environment for the participants. All participants received gifts as a reward for their participation.

The participants were divided into three groups: a baseline group, a CNReader group, and a mature Chinese children’s reading app (anonymous “X” app). The selection of Chinese children’s reading tools for the third group met the following criteria: inclusion of reading materials suitable for the age range of the experimental subjects, display of text materials, and standard reading conditions. There were no significant differences in age, RQ, and SPM scores of the three groups (Table 1).

Procedure. We selected an article for this age group which none of the participants had read before the test. Prior to the formal testing, all participants underwent a baseline reading test. The reading material was printed on A4 paper. The text was in size 3 black KaiTi font, with pinyin in size 5, and a line spacing of 5 mm, adhering to the Chinese national standards for children’s textbook font sizes. All participants completed the reading test

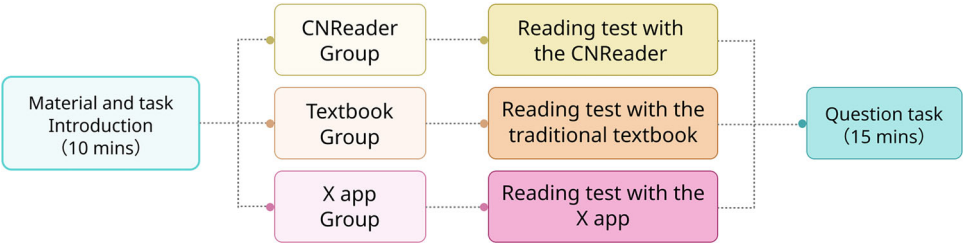


Fig. 4 Procedure of study 1.



Fig. 5 Test scenarios of study 1. **a** Children in the CNReader group read using CNReader; **b** The textbook group read the textbook; **c** The app interface read by group X.

Table 2 The statistical metrics on participants' reading performance.						
Metrics	Baseline group (<i>M</i> ± <i>SD</i>)	CNReader group (<i>M</i> ± <i>SD</i>)	X app group (<i>M</i> ± <i>SD</i>)	Baseline and CNReader <i>Z</i> / <i>p</i>	Baseline and X app <i>Z</i> / <i>p</i>	CNReader and X app <i>Z</i> / <i>p</i>
Finish Time/s	171.48 ± 71.45	103.88 ± 23.03	201.2 ± 35.65	−4.124/0.000***	−2.853/0.004**	−5.745/0.000***
Number of reading errors	12.88 ± 6.36	8.08 ± 3.32	22.96 ± 6.42	−3.007/0.003**	−4.800/0.000***	−6.053/0.000***
Reading comprehension scores	2.48 ± 0.90	3.48 ± 0.50	2.40 ± 0.57	−3.960/0.000***	−0.410/0.682	−5.047/0.000***
Reading rhythm scores	2.60 ± 0.94	3.28 ± 0.53	1.92 ± 0.27	−2.701/0.007**	−3.215/0.001***	−6.324/0.000***

p* < 0.010, *p* < 0.001.

independently. We recorded their reading completion times and conducted a difference analysis on these times. The results indicated no significant differences in reading completion times among the three groups of children ($H = 4.087, p = 0.130$). Then, three groups were informed about the reading task prior to the test. Experimental procedure as shown in Fig. 4. The CNReader group used the CNReader’s single-player practice mode (Fig. 5a), which included instant reading evaluation, user-friendly reading interface, reading punctuation assistance, and image generation after completion of reading. The textbook group used traditional textbook materials (Fig. 5b), and Chinese characters are annotated with pinyin. The X app group utilized the X app to complete the reading task. The interface of the X app features a reading interaction area on the left side, while the right side displays Chinese characters with no options to adjust font size or color (Fig. 5c). The system offers three reading speed options: 0.5×, 1×, and 1.5×, allowing children to choose to follow along. Children conducted the reading test one by one. After completing the reading, we asked the children questions related to

the comprehension of the article. The researchers documented the entire reading process of the children.

Data collection. We collected the children’s reading completion times, the number of reading errors, and invited primary school language teachers as experts to grade their responses to questions and their sense of reading rhythm on a scale of 1–5, with 5 being the highest and 1 the lowest.

Results. The results are shown in Table 2. Before analyzing the differences among the groups, we conducted a normality test on the data. Shapiro–Wilk test showed the data of the three groups were not normally distributed, so we chose the Mann–Whitney *U* test for a non-parametric test.

For the average reading finish time, the Mann–Whitney *U* test showed significant differences between each of the two groups (Table 2). However, when compared to the average reading time of typical children ($M = 62.43$), children using CNReader still took longer.

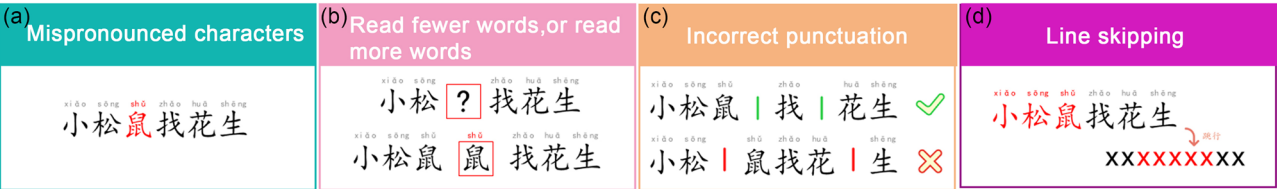


Fig. 6 Reading errors in the textbook group and the X app group. **a** Mispronounced characters; **b** Read fewer words or read more words; **c** Incorrect punctuation; **d** Line skipping.

Table 3 Background information about the participants of study 2.				
Group	N	Age	RQ	SPM
1	11 (F = 5, M = 6)	M = 7.36, SD = 0.64	83.64 (<90)	98.73
2	11 (F = 7, M = 4)	M = 7.82, SD = 0.39	84.27 (<90)	95.82
3	22 (F = 11, M = 11)	M = 7.05, SD = 0.58	85.14 (<90)	97.64

In terms of reading errors and the rhythm scores, it showed significant differences between each of the two groups (Table 2). In expert evaluations of reading comprehension, the baseline group and the CNReader group, the CNReader group and the X app group indicated significant differences, while the baseline group and the X app group showed no significant difference (Table 2). This suggests that the images generated by CNReader based on the text can help children understand the content.

The above data results indicate that users of CNReader outperform those in the baseline group and the X app group. It is evident that the popular reading software X lacks design features tailored for children with dyslexia, such as appropriate background, font, text size, and color settings. Furthermore, the absence of pinyin in X app exacerbates the reading burden for these children (Fig. 5c). Additionally, X app does not offer real-time reading guidance or error correction functionalities, preventing children from receiving immediate feedback during reading.

In the reading tests, we clearly observed the linguistic characteristics of dyslexia in children. As shown in Fig. 5, children in the baseline and X app groups exhibited a higher number of pronunciation errors (Fig. 6a), particularly in the X app group due to the lack of pinyin prompts. They also frequently skipped or added characters while reading (Fig. 6b), a situation less prevalent in the CNReader group thanks to real-time reading guidance. Moreover, incorrect reading rhythm was notably apparent in both the baseline and X app groups (Fig. 6c). Instances of line skipping during reading were also observed in these two groups (Fig. 6d), whereas such issues did not occur in the CNReader group.

Study 2

Participants. In order to evaluate the effectiveness and acceptance of the CNReader AI-paired reading mode, we invited 44 children with mild reading difficulties. They were divided into three groups (Table 3). The tests on RQ and Raven’s Standard Progressive Matrices showed no significant differences among the children in the three groups. All participants and their parents signed informed consent forms.

Procedure. This study used a reading material from the children’s age group that had not been read by the subjects before. The experimental process is shown in Fig. 7.

Group 1: The children first engaged in collaborative reading with a researcher using printed book materials (Fig. 8a). After completing the reading session, they took a 5-min break and then engaged in collaborative reading with the CNReader’s AI assistant.

Group 2: The children first engaged in collaborative reading with the AI assistant, followed by a 5-min break, and then engaged in collaborative reading with a researcher (Fig. 8b).

Group 3: Each child completed the reading task alone using the traditional textbook.

To control experimental variables, we standardized the reading sequence and content between the AI assistant and the researcher. To differentiate between the effects of human–human collaborative reading and human–AI collaborative reading, we independently recorded performance data during each reading session, thus eliminating order effects and mixed effects. After the reading session, children in group 1 and group 2 participated in a semi-structured interview to evaluate the interface friendliness and their cooperative reading willingness towards the two reading methods. The entire experimental process was videotaped.

Data collection. We collected data on the number of reading errors made by children from three groups, and invited language teachers as experts to score the children’s reading fluency on a scale from 1 to 5. Finally, we collected data on the Group 1 and Group 2 children’s perceived levels of interface friendliness and their willingness to engage in cooperative reading for the two different reading methods.

Results. As shown in Table 4, the number of reading errors made by children when cooperating with the AI assistant was fewer than when cooperating with the research associates. Shapiro–Wilk test showed the data of the two groups were not normally distributed. The Wilcoxon signed test indicated a significant difference between the two groups ($Z = -2.542, p = 0.011$). The fluency score of reading with AI cooperation was also higher than that of human–human cooperation, with a significant difference ($Z = -3.289, p = 0.001$). Moreover, the reading interface design of CNReader received positive feedback from children ($M = 4.23, SD = 0.42$). We observed that some children adjusted the font size to their preferred state during reading. Regarding the willingness to engage in cooperative reading, the AI cooperation reading scored significantly higher than the human–human cooperation ($Z = -4.109, p = 0.000$).

In a comparative study on reading with CNReader and independent reading by children, we employed the Mann–Whitney U test (data were not normally distributed) to analyze the differences in the number of reading errors and reading fluency between the two conditions. The results indicated significant differences in both error frequency and fluency (Table 4). This finding not only confirms the effectiveness of CNReader in assisting children with reading difficulties but also highlights its unique value in paired reading. Through interaction with CNReader, children can engage in reading within a safe and supportive environment, enhancing their reading abilities,

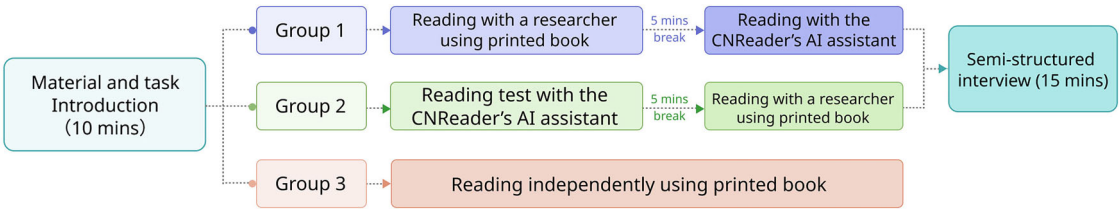


Fig. 7 Procedure of study 2.



Fig. 8 Test scenarios of study 2. **a** The child reads in cooperation with a researcher; **b** The child reads with CNReader’s AI assistant.

Table 4 The statistical metrics on participants’ reading performance are reported.					
Metrics	Human-human cooperative reading (<i>M</i> ± <i>SD</i>)	Human-AI cooperative reading (<i>M</i> ± <i>SD</i>)	Independent reading (<i>M</i> ± <i>SD</i>)	Human-human and human-AI <i>Z/p</i>	Human-AI and independent <i>Z/p</i>
Number of reading errors	8.32 ± 4.28	5.59 ± 1.40	13.27 ± 2.28	−2.542/0.011*	−5.657/0.000***
Reading fluency score	2.27 ± 0.96	3.32 ± 0.55	1.82 ± 0.70	−3.289/0.001***	−5.502/0.000***
Interface friendliness	2.50 ± 0.58	4.23 ± 0.42	—	−4.102/0.001***	—
Cooperative reading willingness	2.05 ± 0.64	4.09 ± 0.67	—	−4.109/0.000***	—

p* < 0.050, **p* < 0.001.

especially in paired reading exercises without the accompaniment of parents or other guardians.

To assess the comparative efficacy of human–human versus human–AI collaborative reading approaches, we analyzed performance metrics within Group 1 and Group 2 participants (Table 5). Findings revealed that children with reading disabilities demonstrated significantly enhanced outcomes through human–AI collaborative reading compared to traditional human–human collaborative interventions.

In interviews, children mentioned that in human–human collaborative reading, both individuals must share the same reading material, making the reading process inconvenient (Fig. 8a). They also felt pressured and worried that reading poorly would embarrass them. These concerns did not exist in the AI-paired context (Fig. 8b). This phenomenon may differ from previous literature that suggested human–human interaction is superior to human–AI interaction in shared reading. It is important to note that the reading scenarios and interaction forms in this study might have influenced these results. Specifically, the design of CNReader aims to enhance personalized experiences, allowing children to adjust interface settings according to their needs, thereby reducing the inconvenience and pressure associated with sharing traditional paper books. Further

analysis revealed that the AI assistant provided a more personalized and autonomous reading experience, enabling children to control their reading pace and manner more freely. This could be a key factor leading to their higher willingness for collaborative reading. In human–human collaboration, the need to coordinate the reading pace and understanding process of two individuals may cause children to feel pressured and inconvenienced, thereby affecting their reading experience and willingness.

Study 3

Participants. We conducted a 1-month follow-up test on the impact of CNReader on children’s reading habits and abilities. A total of 15 children, with an average age of 6.73 years (*SD* = 0.68), diagnosed with mild reading disorders, along with their parents, participated. All children participants completed the RQ tests (*M* = 83.13) and a Raven’s standard progressive matrices test (intellectual ability = normal) prior to the experiment. Researchers provided detailed information to children and their parents about the purpose, procedures, potential risks, and benefits of the study. Informed consent forms were signed by all participants and their parents. All personal information was

Table 5 The statistical metrics on participants' reading performance of Group 1 and Group 2.				
Metrics	Number of reading errors	Reading fluency score	Interface friendliness	Cooperative reading willingness
	Human-human and human-AI Z/p	Human-human and human-AI Z/p	Human-human and human-AI Z/p	Human-human and human-AI Z/p
Group 1	−4.117/0.001***	−3.796/0.001***	−4.220/0.001***	−3.782/0.001***
Group 2	−4.120/0.001***	−4.072/0.001***	−4.220/0.001***	4.202/0.001***
*** <i>p</i> < 0.001.				

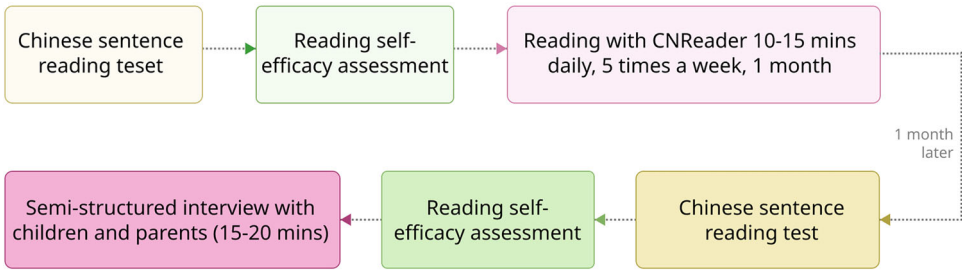


Fig. 9 Procedure of study 3.

anonymized and securely stored. Participants received gifts as compensation for their involvement.

Procedure. The test process is shown in Fig. 9. Prior to the reading exercise, the children underwent a Chinese sentence reading test, which assessed rapid word recognition and reading fluency. This test consisted of 100 sentences, ordered from short to long, utilizing high-frequency Chinese characters and common sentence structures. We recorded the number and accuracy of sentences read by the children within 3 min, as well as the time taken to read all 100 sentences (Lei et al., 2011; Lv and Wan, 2018; Yang and Gong, 1997). Following this, the children underwent a five-point scale assessment of their reading self-efficacy, primarily focusing on reading confidence and interest. The children and their parents were then introduced to CNReader and instructed to install it on their tablets or computers at home. During the 1-month trial period, the children were required to use CNReader for 10–15 min daily, five times a week, under the supervision of their parents. We provided 30 reading materials suitable for their age group, which could be directly accessed through CNReader. One month later, we once again conducted the Chinese sentence reading test and the reading self-efficacy assessment on the children, and carried out a 15–20 min semi-structured interview with each child and their parent. To gain more detailed user experience data, particularly regarding the usability and engagement of CNReader from both children and parents, we incorporated additional specific questions into our semi-structured interview process. These questions included, but were not limited to: What features of CNReader do children most and least enjoy? Has the system increased children’s interest and confidence in reading? Have parents observed any changes in their children’s reading habits or performance? These questions have enabled us to gain a comprehensive understanding of how users interact with the tool and their overall satisfaction with it.

Data collection. We collected data on children’s Chinese sentence reading tests before and after the experiment, self-efficacy evaluation data on reading. Through semi-structured interviews, we not only collected evaluations from children and parents regarding the user experience and functional design of CNReader, but also paid particular attention to user engagement. We

analyzed the features that children liked and disliked the most, as well as any changes in reading habits or behaviors observed by parents.

Results. As shown in Fig. 10, after 1 month of practice, improvements were observed among the 15 children. In the Chinese sentence reading test, the children were able to read more sentences within 3 min ($M_{pre} = 8.20$ SD = 3.58, $M_{post} = 14.87$ SD = 1.71), and the Wilcoxon signed-rank test indicated a significant difference between the two measurements ($Z = -3.125$, $p = 0.001$). The number of correctly read sentences within 3 min also showed a significant improvement ($M_{pre} = 8.20$ SD = 3.58, $M_{post} = 14.87$ SD = 1.71, $Z = -3.432$, $p = 0.001$). Furthermore, the time taken to read 100 sentences significantly decreased ($M_{pre} = 78.80$ SD = 17.91, $M_{post} = 55.07$ SD = 15.42, $Z = 3.409$, $p = 0.000$). In terms of reading self-efficacy, the children’s reading confidence significantly increased ($M_{pre} = 2.07$ SD = 0.57, $M_{post} = 3.80$ SD = 0.54, $Z = -3.473$, $p = 0.001$), and their interest in reading was significantly enhanced ($M_{pre} = 2.13$ SD = 0.62, $M_{post} = 4.47$ SD = 0.50, $Z = -3.501$, $p = 0.000$).

During the semi-structured interviews, we further explored the experiences and perceptions of children and parents regarding the specific functions and design of CNReader.

Usability. Several children (e.g., Child 3, Child 12, Child 14) mentioned that they appreciated the feature allowing them to adjust the font size, which made reading more comfortable. Children aged 5 and 9 noted that the immediate feedback function was particularly helpful, as it enabled instant identification and correction of errors, thereby enhancing their learning outcomes. Most respondents indicated that the AI-paired reading mode was the most popular feature. This mode provided personalized guidance and engagingly increased their interest in reading. Parents generally found CNReader to offer a wealth of reading materials, which could be converted into dyslexia-friendly text formats, making it highly practical. The AI-paired reading mode was viewed by parents as a significant advantage, as it somewhat replaced the need for a human reading companion, alleviating parental guilt associated with being too busy to read with their children. However, C6 reported that the system’s voice feedback was occasionally too fast, making it difficult for him to

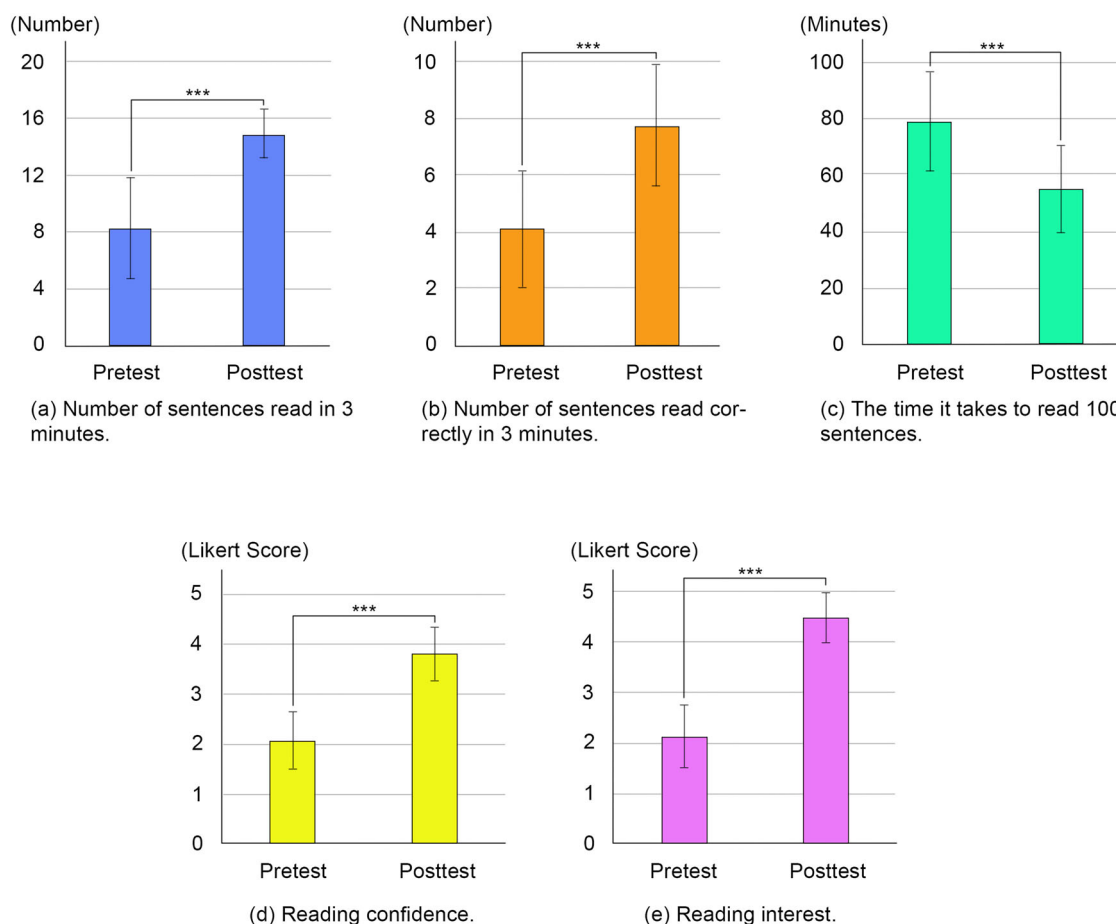


Fig. 10 Children practice using CNReader for 1 month. a Number of sentences read in 3 min; **b** Number of sentences read correctly in 3 min; **c** The time it takes to read 100 sentences; **d** Reading confidence; **e** Reading interest.

keep pace. Additionally, C2 and C10 expressed a desire for more interactive elements, such as mini-games or reading challenges, to make the experience more engaging. Some parents hoped for more customization options, such as the ability to select reading materials of specific difficulty levels or to create tailored reading plans.

Engagement. CNReader has achieved significant improvement in enhancing children's reading skills and confidence. Through its personalized features, instant feedback mechanism, and gamified design, CNReader effectively increases children's engagement. Most children report that they enjoy interacting with CNReader daily, and in interviews, all children expressed their willingness to use CNReader again. Based on observations and feedback from parents, CNReader's gamified design (e.g., unlocking images within articles through repeated reading) greatly enhances children's motivation and engagement in reading. Several parents have reported that their children are willing to read repeatedly to unlock more images. This design effectively combines learning with entertainment, thereby enhancing children's interest in reading.

Reading performance. As the experimental results indicate, the children demonstrated significant progress in the Chinese sentence reading test. Interviews further corroborate this finding; Child 4 mentioned that the instant feedback feature provided by CNReader helped him immediately identify and correct reading errors. This not only improved his reading accuracy but also

increased his interest and confidence in learning. Many parents reported substantial improvements in their children's rapid character recognition and reading fluency. For example, the mother of Child 13 remarked, "My child used to get stuck on some simple characters, but now he can recognize and understand them more quickly, which has boosted his reading confidence".

Reading habits. In interviews, parents reported an increase in their children's reading frequency and duration over the past month. During the period of using CNReader, their children were able to maintain the daily reading requirement of 10–15 min and gradually developed a fixed reading schedule. For instance, the mother of Child 5 mentioned, "My child rarely read voluntarily in the past, but now uses CNReader every day and has become more self-disciplined." In addition to having a set reading time, many children also choose a quieter and more comfortable reading environment. The father of Child 7 noted, "We have a small reading corner at home, and the children now prefer to use CNReader there, which has greatly improved their attention and reading effectiveness".

Discussion

Cognitive basis for enhancing reading abilities with CNReader.

In this study, we observed significant improvements in reading accuracy, fluency, and comprehension with the use of CNReader, which are closely linked to the specific cognitive challenges faced

by children with developmental dyslexia. These children typically exhibit deficits in phonological awareness, morphological awareness, and orthographic awareness, which directly impede their reading ability development (Huang et al., 2023). CNReader effectively aids children in decoding and recognizing Chinese characters through color-coded visual cues and structured text layouts, thereby enhancing their orthographic awareness.

Simultaneously, CNReader's auxiliary sentence segmentation function provides guidance on reading rhythm, assisting in the improvement of children's phonological and morphological awareness. Studies have shown that improved phonological awareness helps children better understand the sound structure of language, while enhanced morphological awareness facilitates the recognition and use of the smallest units of meaning in language (Li et al., 2023). By strengthening these key cognitive areas, CNReader effectively improves children's reading fluency and comprehension.

Additionally, the AI-paired reading mode offered by CNReader cultivates sustained attention in children during extended reading activities by providing personalized practice with varying text complexities. Sustained attention is central to fluent reading and information-processing speed, and many children with developmental dyslexia lack this ability. In our study, the introduction of the AI-paired reading mode significantly improved the children's reading precision and fluency, indicating its potential in helping to overcome attention deficits.

By integrating these cognitive foundations with the training effects, CNReader not only provides support for children with developmental dyslexia but also offers educators and parents a cognitive framework to understand its efficacy. This framework highlights the critical role of comprehensive interventions targeting phonological, morphological, and orthographic awareness in improving reading skills, while also providing a theoretical basis for future research exploring the link between cognitive interventions and actual reading skill enhancements.

The role of AI voice assistants in children's paired reading. Our tests indicate that children tend to prefer AI as their reading companion during shared reading sessions. In semi-structured interviews, these children mentioned that AI voice assistants provide a stress-free learning environment, allowing them to learn at their own pace and comfort. The real-time feedback function offered by these assistants helps them understand their mistakes and provides corrective strategies, thereby improving their reading skills. Previous research also supports this finding. For instance, some studies have shown that AI-assisted learning can enhance the reading comprehension and fluency of children with dyslexia (Yang, 2022; Bhushan et al., 2024). These studies point out that AI technology can personalize learning content, enabling children to learn in a more targeted environment.

Furthermore, parents have expressed a desire for CNReader to offer more comprehensive language skills training in the future to help children improve their listening, speaking, and writing abilities. This demand indicates that expanding the functionality of AI assistants is an essential direction for future development. Moreover, previous literature has mentioned the significant potential of AI in multimodal learning, which can enhance learning outcomes (Rajagopal et al., 2023).

While some parents acknowledge the potential of AI assistants in education, they also worry about children becoming overly reliant on AI assistants and CNReader, which might affect their reading abilities in real-world settings. Existing research (Han et al., 2024) also points out that over-reliance on technology may have negative impacts on children's independent learning abilities

and social adaptability. Therefore, future research needs to explore how to balance AI-assisted learning with traditional teaching methods, ensuring that children can effectively develop their reading abilities in different learning environments.

It remains unclear whether using AI assistants for paired reading affects children's reading abilities and social adaptation skills in real-world environments. This is an important area that requires further research. By comparing the long-term effects of AI-assisted learning and traditional learning methods, future studies can provide educators and parents with more detailed guidance, enabling them to make more informed choices about the most suitable educational tools and methods. In summary, the role and significance of AI in training the reading skills of children with dyslexia are not only reflected in personalized learning content and real-time feedback but also require future research to explore its long-term educational impacts and the balance point in practical applications.

Expansion of CNReader to dyslexic populations in other languages. Given that developmental dyslexia manifests unique linguistic cognition and information-processing deficits in different languages, the fundamental principles of CNReader can be adapted for dyslexic readers of other languages, such as English, Spanish, or Japanese. These languages face similar challenges regarding reading fluency and accuracy. Firstly, the user-friendly visual text presentation of CNReader can be customized according to the orthographic and phonological features of other languages. For example, in alphabetic languages such as English or Spanish, the tool can highlight phoneme-grapheme correspondences, provide syllable segmentation, and employ color-coding strategies to enhance decoding skills. Previous research has shown that phoneme-grapheme correspondences and syllable segmentation significantly impact reading speed and accuracy for dyslexic individuals (Meng, 2018).

Secondly, personalized reading guidance and visual cues, such as text highlighting and segmentation, are beneficial in other language learning environments. Specifically, for languages with complex structures and where prosody and rhythm are crucial for comprehension, such as Italian or French, the CNReader approach can guide phrase pauses and provide feedback on rhythm and prosody, aiding in improved reading fluency. Existing research indicates that dyslexic individuals often require visual and auditory cues to enhance reading efficacy when dealing with the prosody and rhythm of these languages (Meng, 2018).

Thirdly, CNReader's AI-paired reading support is adaptive and interactive. The AI can ensure proper pronunciation, provide instant feedback on reading errors, simulate paired reading experiences, and encourage repeated reading practice, thereby increasing engagement and motivation. Current literature shows that AI applications in language learning have achieved significant results, with their interactivity and prompt feedback being indispensable for improved learning outcomes (Amonova et al., 2024; Seow, 2023). Future research could focus on the adaptation and validation of CNReader in different linguistic environments, incorporating insights from cross-language studies on dyslexia and human-computer interaction. Particularly, through experimental design and data analysis, validating the efficacy of CNReader across different language groups will provide valuable data support for cross-linguistic dyslexia research.

In conclusion, CNReader has demonstrated significant effectiveness among Chinese dyslexic readers, and its methods and technological framework hold broad applicability for dyslexic populations in other languages as well. With continuous

technological advancements and deeper cross-linguistic research, CNReader is poised to become an effective tool for dyslexia intervention on a global scale.

Limitations and future work. We have reported on a study of an intelligent reading practice tool designed for children with developmental Chinese dyslexia. We acknowledge some limitations of the study and outline future work to address them.

Firstly, in this study, although we have designed a user-friendly CNReader interface that incorporates dyslexia-friendly fonts and colors for English, and fonts, font sizes, and line spacing that are suitable for Chinese, we have not conducted thorough testing to assess the actual effectiveness of these combined elements. Therefore, our interface design may have certain limitations. In the future, we will continue to delve deeper into this area to refine the CNReader interface design. We will conduct rigorous user testing with dyslexic children to collect comprehensive feedback and employ quantitative measures to evaluate the usability and effectiveness of the interface elements.

Secondly, while our experimental studies assessed both immediate effects and outcomes following a 1-month training period with CNReader, dyslexia interventions require long-term evaluation. Therefore, short-term experimental results may inadequately reflect the tool's longitudinal effects. A significant limitation in Study 3 was the absence of immediate post-intervention assessment, with only 1-month follow-up data collected. Without a control group and immediate post-test measurements, it remains unclear whether the observed improvements can be attributed solely to the CNReader intervention or whether they resulted from other factors such as natural development, concurrent educational activities, or testing effects. Future research will include larger participant samples and extended training assessment periods. Additionally, we will implement longitudinal research designs to elucidate the sustained efficacy of CNReader across extended timeframes and varied contextual conditions.

Thirdly, inherent potential biases in our research design and sample selection may affect the generalizability of our research results. For example, our research participants were limited to a specific age group and geographical region. Future research should include a more diverse sample, encompassing different socioeconomic backgrounds, regions, and age groups, to enhance the external validity of the research results. Additionally, we need to consider and control for other confounding variables that may influence the results, such as prior exposure to reading interventions and varying levels of parental support.

Lastly, our CNReader currently only supports reading training functions. However, dyslexia in Chinese is complex, with significant individual differences, requiring targeted intervention methods. Therefore, the functionality of CNReader may have certain limitations and may not fully meet the needs of individuals with dyslexia. In the future, we will further explore more aspects of dyslexia in Chinese and propose more targeted feature designs. This includes optimizing adaptive learning algorithms to dynamically adjust training content based on individual progress and difficulties, and integrating multimodal approaches with auditory and tactile feedback.

To address the limitations and further enhance and validate the tool, our future work will focus on the following specific research questions and methodologies:

Usability and effectiveness study. Conduct comprehensive usability studies on a larger, more diverse group of children with dyslexia. This study will include qualitative interviews and

quantitative measures such as eye-tracking and task performance metrics to evaluate the actual effectiveness of different interface elements.

Long-term impact assessment. Implement a longitudinal study, carrying out multiple checkpoints over an extended period (e.g., 6 months to a year) to assess the sustained impact of CNReader on reading skills. This will involve regular assessments and progress tracking to observe long-term improvements and retention.

Generalization and bias mitigation. Expand the participant pool to include a broader demographic range and control for potential confounding variables. We will use stratified sampling techniques to ensure representation from different socioeconomic statuses, regions, and age groups, thereby enhancing the generalizability of our findings.

Adaptive and multimodal features. Develop and test adaptive algorithms for personalized reading exercises based on real-time performance data. Additionally, integrate multimodal feedback (visual, auditory, and tactile) to cater to different learning preferences and needs, providing a more holistic approach to reading intervention.

User customization options. Allow parents and educators to customize AI reading sessions based on specific goals and preferences. This could include selecting types of feedback, adjusting levels of interactivity, or choosing topics and characters that interest the child.

By addressing these issues, our aim is to overcome current limitations and provide a more robust and effective tool for children with developmental dyslexia in reading Chinese, ultimately contributing to better educational outcomes and quality of life.

Conclusion

This study presents a reading training tool named CNReader, specifically designed for children with mild Chinese dyslexia. The tool features a user-friendly visual interface and offers three reading modes: single-player reading practice mode, AI-paired reading mode, and reading challenge mode. It can assess children's reading level in real-time and provide personalized reading guidance and smart reading companionship. We validated the effectiveness of CNReader in both single-player reading practice mode and AI-paired reading mode, and explored its impact on children's medium- and long-term reading training. The results showed that after using CNReader, children's reading accuracy, fluency, interest in reading, and self-confidence have all improved. Moreover, children demonstrated a high level of acceptance towards collaborative reading with AI. However, there are still areas for improvement in the design of CNReader and its study setup. We look forward to constantly optimizing it to better assist children with Chinese dyslexia in their reading training.

Data availability

The data utilized in this study are available upon request from the corresponding author. All data have been anonymized and can be made accessible for academic research purposes, subject to reasonable requests. Data sharing will comply with relevant confidentiality agreements and ethical guidelines to safeguard the privacy rights of minor participants.

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Author contributions

Lijuan Liu: conceptualized the study and designed the overall research methodology. Led the development of the CNReader tool and supervised the experiments. Designed the AI reading support system, ensuring adaptation to individual user needs. Drafted the paper,

including the abstract, methodology, and results. Tuo Fang: conceptualized the study and provided the experimental sites and funding. Enmao Liu: provided expertise on the cognitive and academic challenges faced by children with developmental dyslexia, contributing to the design of the reading guidance. Shang Shi: conducted the statistical analysis of the collected data, interpreting the results from the three experiments. Shuo Zhai: organized and managed the participation of the 57 children with mild dyslexia, including obtaining consent and scheduling sessions. Yang Chen: worked on the user interface design of CNReader, focusing on creating a child-friendly and engaging visual text presentation. Lingyan Zhang: implemented and optimized the AI-paired reading feature, ensuring it could effectively assist users in improving their reading skills. Yan Shi: oversaw the study timeline and resource allocation, ensuring the project stayed on track and within budget. Cheng Yao: drafted the paper and coordinated feedback from all contributors to finalize the manuscript.

Competing interests

The authors declare no competing interests.

Ethical approval

The study "CNReader: a reading practice tool designed for Chinese children with developmental dyslexia" received ethics exemption qualification on May 7, 2023. The following constitutes the basis for exemption from ethical review. (1) Informed Consent: This study obtained written informed consent from participants between May 20 and May 25, 2023. The experimental subjects were children with mild reading disabilities and their parents. All age-appropriate children signed the Informed Consent Form, accompanied by their parents. The document explicitly stated the experimental objectives, content, and the scope of data utilization. Participants retained the right to withdraw at any time. The original consent forms were archived by the research team, with copies provided to the participating families. (2) Content of Experiments: The experimental tests involve reading exercises from standard educational materials, including reading tests of textbook passages and routine sentence tests (please refer to the experimental setup for details). (3) Design and Evaluation: The experimental design, test content, and data analysis were evaluated before the experiment by experienced primary school Chinese language teachers and parents. The reading exercises mirror those that commonly occur in daily classroom settings. (4) Exemption Criteria in China: According to the "Measures for Ethical Review of Life Sciences and Medical Research Involving Human Subjects" (issued by the National Health Commission of China, Ministry of Education, Ministry of Science and Technology, and National Administration of Traditional Chinese Medicine, implemented on February 18, 2023), Article 32 stipulates that ethical review may be exempted for research that does not cause harm to the human body, does not involve sensitive personal information or commercial interests, in order to reduce unnecessary burden on researchers: Research utilizing legally obtained public data, or data generated through observation without interfering with public behavior; Research conducted using anonymized information and data. Based on the above considerations, this study was granted exemption from requiring ethics approval.

Informed consent

In this study, all participants and their parents provided informed consent. The details are as follows: Prior to the commencement of the study, researchers provided comprehensive explanations to all child participants and their parents regarding the study objectives, procedures, potential risks, and benefits. All informed consent was obtained in written form before the study began, jointly signed by the participating children and their legal guardians (parents). The scope of informed consent included: participation in research activities, collection and utilization of personal data, and consent for anonymizing personal information. The consent form explicitly stated that participation was entirely voluntary, and participants could withdraw at any time without any negative consequences. As this study involved minors (first-grade children with mild reading disabilities), we specifically ensured the following. (1) Recruitment of participants was facilitated through school Chinese language teachers and homeroom teachers. (2) Age-appropriate language was used to explain the research content to children. (3) Consent was obtained from the children themselves as well as written permission from their legal guardians (parents). (4) School counselors and psychologists were available to provide support for children experiencing discomfort or distress. All participants' personal information was anonymized and securely stored to protect their privacy and data security. Participants received gifts as compensation for their participation upon completion of the study.

Additional information

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